

Histogram

A Histogram is a graphical representation of the distribution of numerical data. It is an estimate of the probability distribution of a continuous Variable and is used to visualize the shape, central tendency, and variability of a dataset.

Agus = {11, 12, 14, 18, 24, 26, 30, 35,
36, 37, 40, 41, 48, 43, 50}

↓
Histogram

① NO. of bins $\Rightarrow 10$

$\frac{50}{10} = 5 \Rightarrow$ bin size

Bins $\rightarrow$ 0-5

5-10

10-15

15-20

20-25

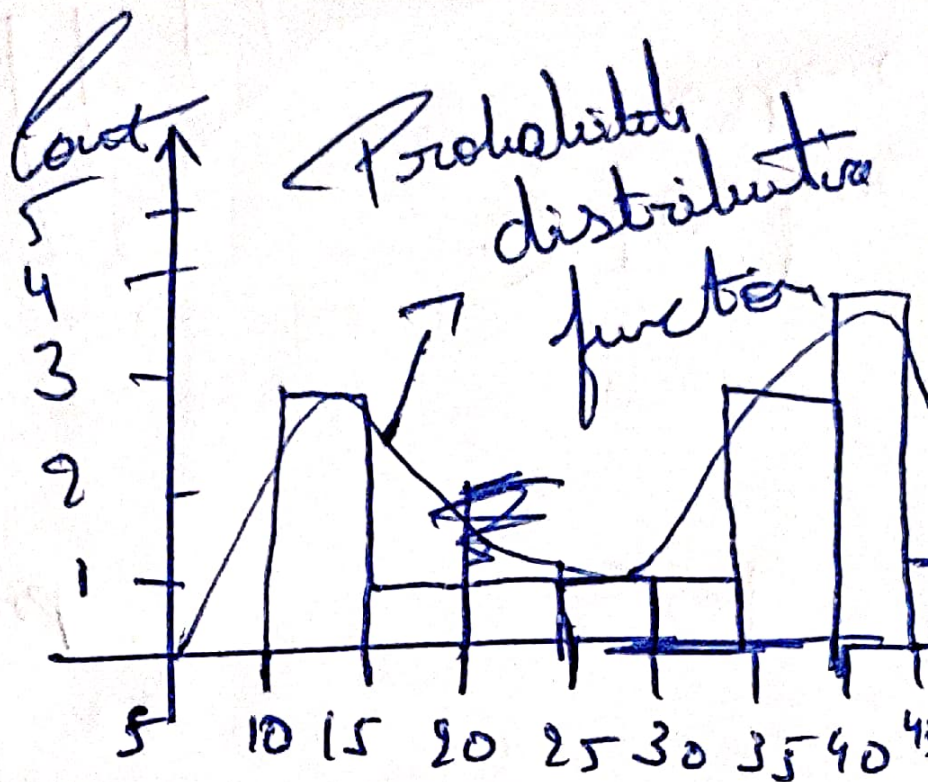
25-30

30-35

35-40

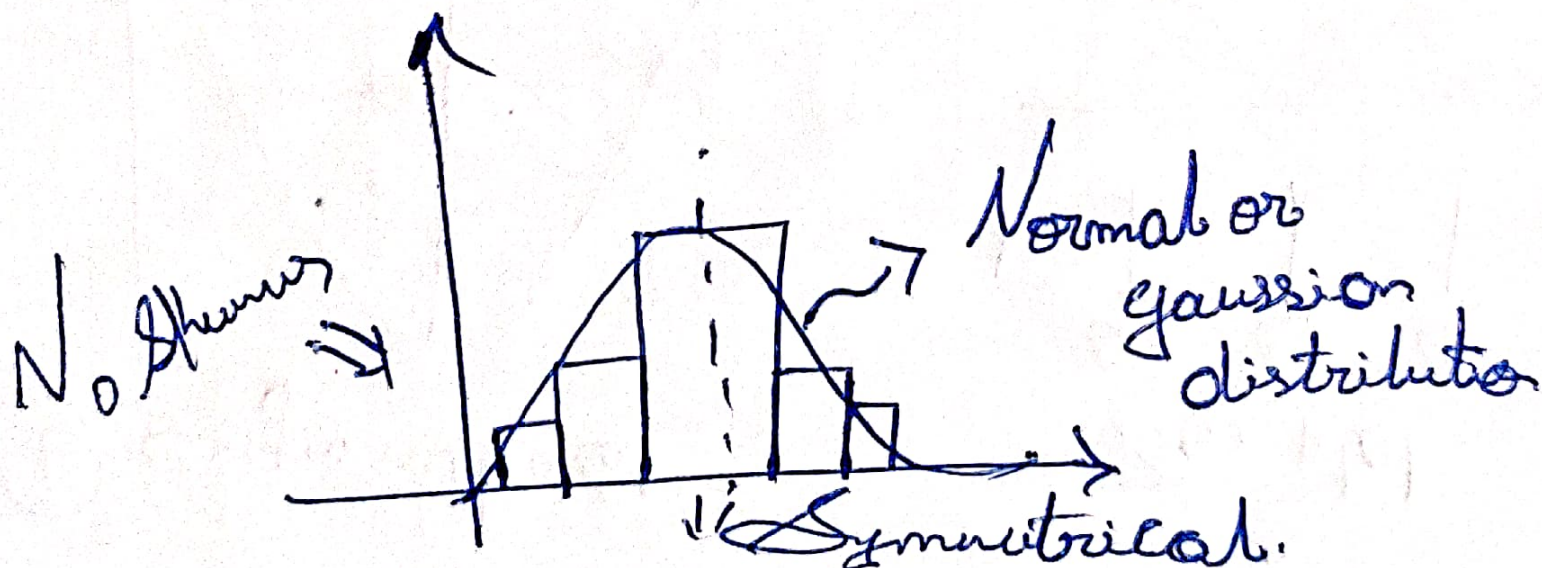
40-45

45-50

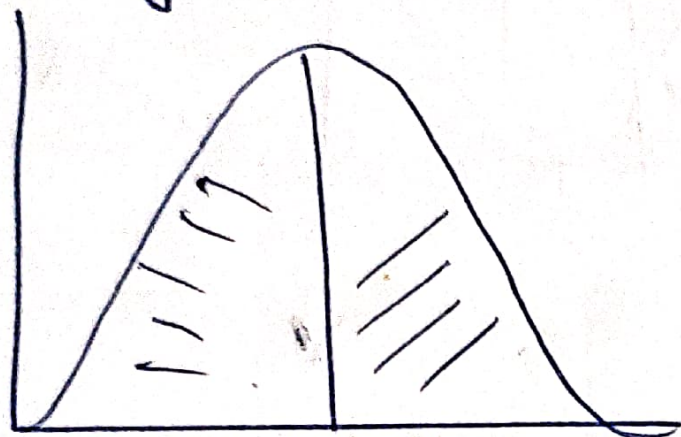


Kernel density Estimator

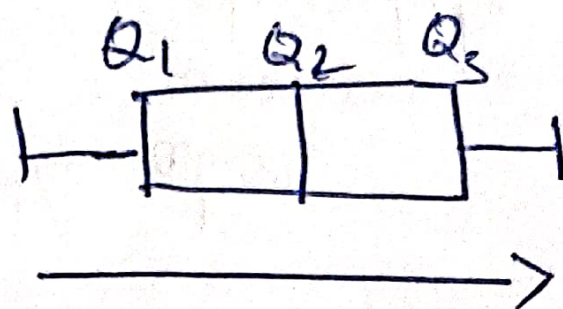
Skewness



① Symmetrical



Box plot

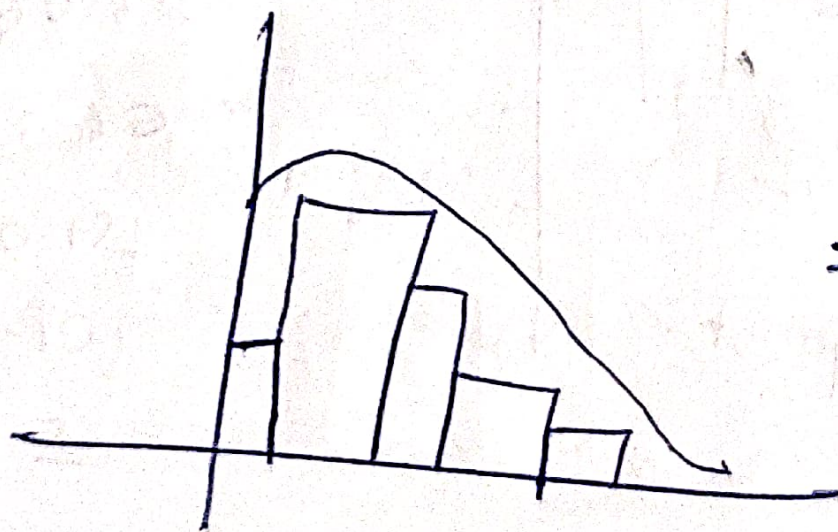


$$Q_3 - Q_1 \approx Q_2 - Q_1$$

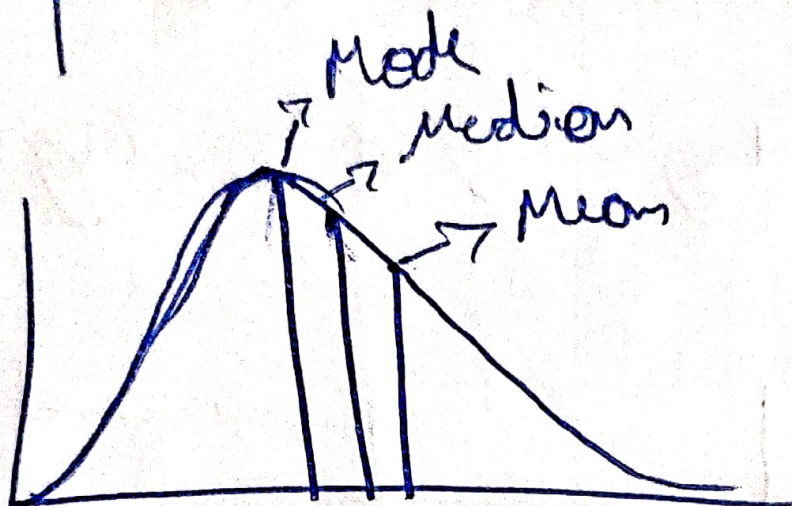
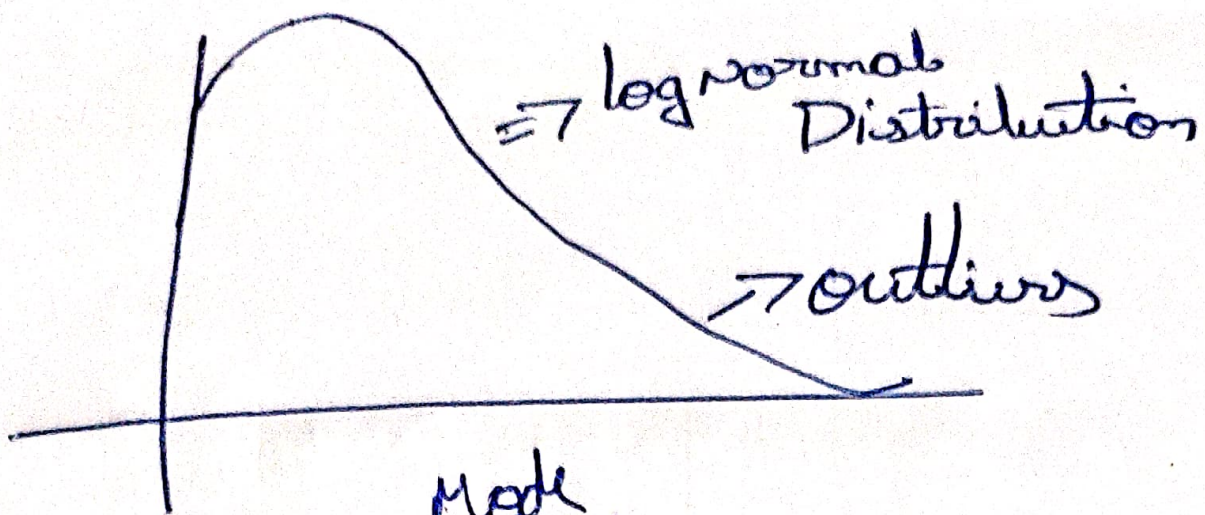
The Mean, Median, Mode
are perfectly at the center

$$\text{Mean} = \text{Median} = \text{Mode}$$

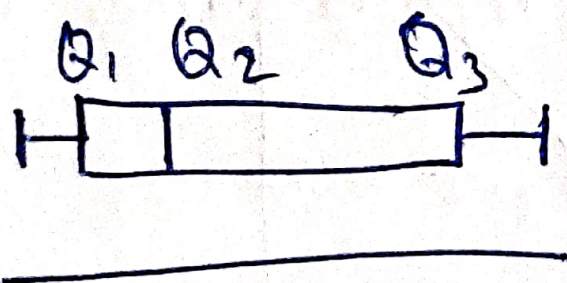
② Right Skewed



$\Rightarrow$ Positive
Skew

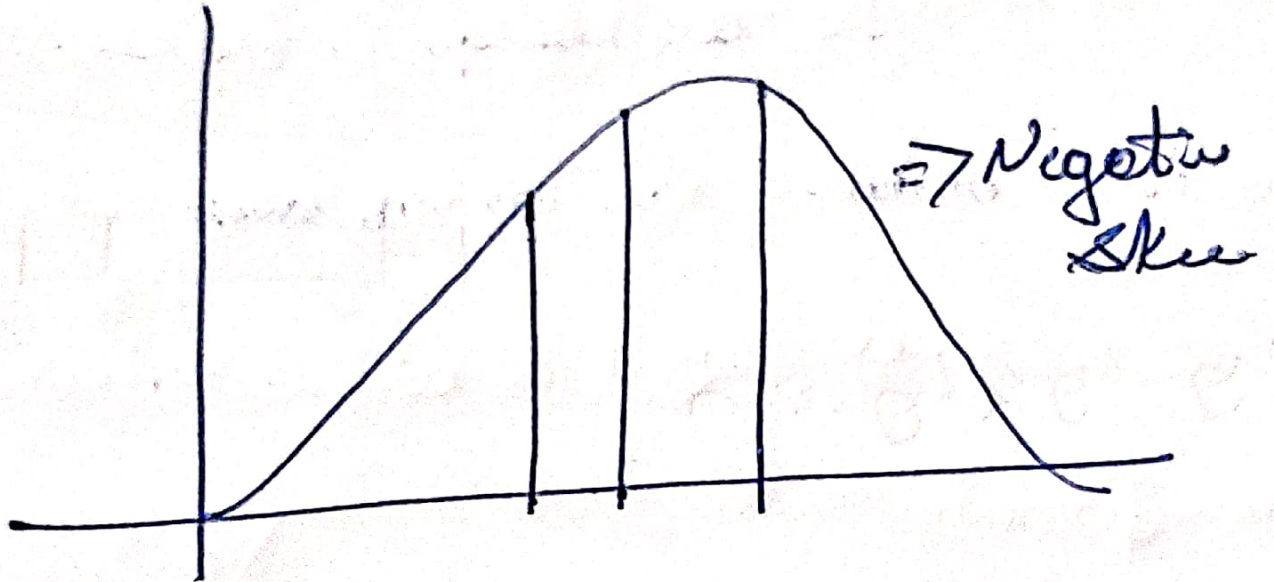


$\text{Mean} > \text{median} > \text{mode}$

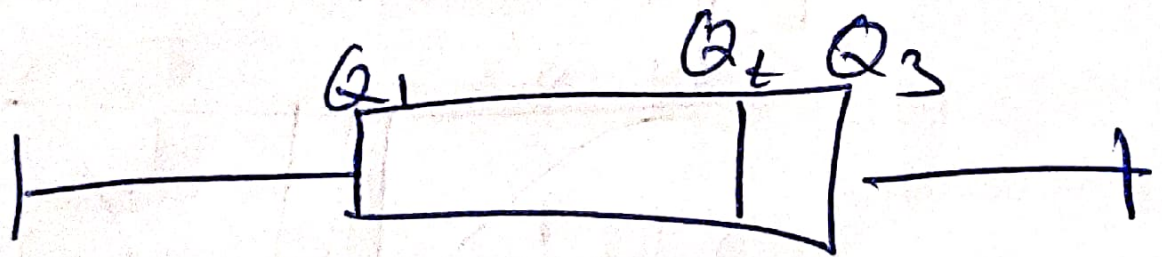


$$Q_3 - Q_2 > Q_2 - Q_1$$

③ Left Skewed



$$\text{Mean} \leq \text{Median} \leq \text{mode}$$



$$Q_2 - Q_1 > Q_3 - Q_2$$